Project Title
TextHack Intelligent Language Query System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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Submitted by
JAKKE VIVEK (2520030180)
KUMMARI PARTHEEV PRANEETH (2520030197)
KARI PRUDHVI SAI (2520030286)

Team Number: 7
Section: 5

Under the guidance of

MS D SWATHI

ASSISTANT PROFESSOR, CSE

Hyderabad-500090, Telangana, India.

ABSTRACT

TextHack+ is an advanced text analysis engine written in Java that is able to process large text datasets by implementing hand-built and highly complex algorithms. The system offers support for multiple querying types like exact pattern matching, fuzzy search with edit distance, document similarity that uses suffix structures, citation-flow analysis based on the max-flow algorithms, task scheduling illustrations for solving NP-hard problems, and probabilistic primality testing.

Project implementation relies on a layered architecture starting from corpus ingestion, followed by manual algorithm engine, query services, and user interface. One of the major restrictions in the engine is that it is done from scratch without the use of java.util.* libraries which calls for custom implementation of all data structures. This aspect showcases deep understanding of algorithms and proves to be one of the biggest reason why more complex structures are essential for performing efficient text processing.

By combining algorithms and their implementations into a single tool, TextHack+ is a practical application that demonstrates theoretical studies of algorithms. Each query is tied to an algorithm family in such system. Also, the system displays both results and algorithm information which makes it highly applicable as an educational tool, benchmarking tool, or research tool. The system is multi-lingual and works Mostly well in case of Indian-language Wikipedia corpora, On top of which it offers an option to do searches in the areas of medicine and domain applications with the help of TextHack+.

Signature of the Guide